

IP-SAKTI SAHAYAK

Family 9 — Piperaceae / Betel (Piper betle)

Objective: Build a TKDL-only Piperaceae cluster around Betel leaf so IP-SAKTI can retrieve closely related Piper plants and rank them as potential alternatives for further research when the user's formulation encounters relevant prior-art or IP risk.

Strict TKDL rule: Every candidate below is explicitly present in the official TKDL plant-name index. No outside species have been added to force the count. Similarity values are prototype ranking values and do not establish equivalence or legal clearance.

1. Anchor / main plant

Betel leaf — Piper betle Linn.

TKDL Ayurvedic terminology includes Nigavall / Cavya-bheda; Unani entry includes Tambol/Paan; Tamil names include Vetrilai and Thamboolavalli. **Primary plant part for this cluster:** leaf (the traditional betel-leaf use represented in the public TKDL examination evidence).

Important distinction: TKDL's public plant index does not provide a single "main active component" field for Piper betle. IP-SAKTI should therefore store "plant part / traditional ingredient role" separately from any later phytochemical data.

2. Related TKDL species — 5 candidates

TKDL plant	Relationship	Similarity*	Evidence / model use
Piper longum Linn.	Same genus; Piper	VERY HIGH — 0.94	Strong TKDL formulation/prior-art evidence; Pippali
Piper nigrum Linn.	Same genus; Piper	VERY HIGH — 0.94	Strong TKDL formulation/prior-art evidence; Black Pepper
Piper cubeba Linn. f.	Same genus; Piper	HIGH — 0.86	Official TKDL index; formulation match should be retrieved
Piper retrofractum Vahl Syn. P. chaba Hunter	Same genus; Piper	HIGH — 0.88	TKDL formulation/examination evidence; Cavya
Piper brachystachyum	Same genus; Piper	HIGH — 0.84	Official TKDL index; evidence depth to be enriched

*Prototype similarity ranking only. Same-genus proximity is a screening signal; exact use/formulation/plant-part evidence must determine the final score.

3. Official TKDL plant-index verification

Official TKDL plant-name index: <https://www.tkdل.res.in/tkdل/langdefault/common/Utility/Keyword/Plant-Name.asp>

The official TKDL index lists Piper betle, Piper cubeba, Piper longum, Piper nigrum, Piper retrofractum, Piper brachystachyum and additional Piper entries such as P. album, P. attenuatum, P. boehmeriaefolium, P. peepuloides, P. schmidtii, P. sylvaticum, P. thomsoni, P. trichostachyon, P. trioicum and P. wallichii. This gives IP-SAKTI a much larger Piper-only expansion pool later.

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4. TKDL evidence — Betel leaf / Piper betle

20100203117: The official TKDL examiner index identifies the application "Anti-adipogenic compositions containing piper betle and dolichos biflorus" and describes the TKDL use of Piper betle and Dolichos biflorus for obesity and urinary disorders. The examiner decided to reject specified claims in 2015.

Official TKDL examiner index: <https://www.tkdل.res.in/tkdل/langdefault/common/ExaminerReport.asp?homepage=sub>

201721045968: The official Patent Office examination report discusses a claimed tobacco-substitute product comprising betel leaf (Piper betle), lemongrass, Pippali (Piper longum) and green chillies. The report states that TKDL references included traditional betel use as mouth freshener/cleanser and anti-inflammatory use, and it found the relevant claims lacking inventive step in light of prior art.

Official TKDL examination PDF: https://www.tkdل.res.in/tkdل/Outcomes/201721045968_Examinationreport.pdf

5. Related-species evidence

Piper longum: TKDL examination material repeatedly records Pippali in multi-ingredient formulations. In 201721045968, Piper longum is part of the claimed tobacco-substitute composition with Piper betle; the report also identifies traditional use of Pippali in a formulation for inflammation. This makes it the strongest related candidate for formulation-similarity retrieval.

Piper nigrum: TKDL public examination records repeatedly pair Black Pepper with Piper longum and other plants in traditional formulations. For example, 1922/DEL/2005 cites Piper nigrum and Piper longum in prior-art formulations for bronchial asthma, cough/bronchitis, pneumonia and related respiratory conditions.

Official TKDL record: https://www.tkdل.res.in/tkdل/LangFrench/common/Examinarreport/reports/1922_DEL_2005.asp

Piper retrofractum: Official TKDL examination material identifies Piper retrofractum (syn. P. chaba Hunter) as Cavya and records it as an ingredient in a traditional formulation. This provides stronger evidence than a plant-index-only candidate.

Official TKDL PDF: https://tkdل.res.in/tkdل/LangFrench/Common/Examinarreport/pdf_files1/1865-DEL-2006.pdf

Piper cubeba: The official TKDL plant index lists Piper cubeba as Ka■kola/Kababchini and provides the traditional/Unani terminology. For this first-pass cluster, it is retained as a same-genus candidate; formulation-specific evidence should be linked before a high-confidence alternative is shown.

Piper brachystachyum: The official TKDL index lists it as a Piper species with the terminology Cavya and common names Pahari Pipar/Pahari Pan. Because the public index alone does not establish a formulation match to Piper betle, it receives a lower evidence level until the corresponding TKDL record is extracted.

6. Alternative-ranking workflow

Example: User enters a Piper betle-based formulation and IP-SAKTI detects relevant prior-art. The engine identifies Piper betle → retrieves same-genus TKDL candidates → compares plant part + traditional indication + formulation role + exact TKDL evidence → ranks candidates.

Example output: “Piper longum — Very High similarity candidate. Same genus as Piper betle and directly co-occurs with betel leaf in a TKDL-linked patent examination record. Investigate further; this does not establish that it is an equivalent substitute or that the resulting formulation is legally clear.”

7. Model-ready dataset fields

family | anchor_plant | candidate_plant | botanical_name | TKDL_name/synonyms | plant_part | traditional_use | formulation | formulation_role | source_text | TKDL_case_id | official_TKDL_url | similarity_score | similarity_reasons | patent_IP_flag | evidence_level | verification_status

8. AI / legal guardrails

- Same genus does not mean interchangeable.
- Same genus does not establish freedom from patent infringement.
- Do not call any candidate “legally safe” without a separate claim-level patent analysis.
- Preserve the exact botanical identity and synonyms appearing in TKDL.
- Treat co-occurrence in a formulation as evidence for similarity retrieval, not proof of efficacy equivalence.
- Show the exact TKDL record behind every high-confidence recommendation.
- Recommended UI wording: **“Potential alternative candidate for further research — TKDL similarity signal detected; botanical, formulation, efficacy and patent/legal verification required.”**

9. Dataset expansion opportunity

Piper is particularly useful for the IP-SAKTI prototype because the official TKDL plant index exposes many Piper entries. After the five-node MVP cluster is validated, the team can expand to *P. album*, *P. attenuatum*, *P. boehmeriaefolium*, *P. peepuloides*, *P. schmidtii*, *P. sylvaticum*, *P. thomsoni*, *P. trichostachyon*, *P. trioicum* and *P. wallichii*, while retaining an evidence-level field so index-only species are not confused with formulation-verified species. ■cite■turn0search1■